1.1 Systematic Estimation

The 20th century physicist Enrico Fermi was known for performing 'back of an envelope' calculations which often gave surprisingly reliable and *accurate* results. Calculations of this type are therefore sometimes called *'Fermi estimates'*. The goal of these *systematic estimates* is to quickly get a sense of the *scale* of numerical answers to complex or even impossible questions such as:

Questions

- *How many times might a teacher take a class register in their career?*
- *How much on average does a large supermarket spend on staff wages each week?*
- *How many miles are driven by cars on Scottish roads in a month?*

These calculations might involve using a mixture of *approximated*, *estimated* and *exact* numbers. For example:

- There are **exactly** 60 seconds in one minute.
- There are **approximately** 30 days on average in a month.
- We might **estimate** the mass of a typical family car to be 1000 kilograms.

It is important to communicate clearly any **assumptions** made in reaching your answer, such as the *estimation* above.

Example

Problem:
A secondary school in a city centre in Scotland has approximately 800 pupils.
Some buy their lunch in the dining hall and some buy their lunch from shops or bring their lunch from home.
Estimate the total amount spent on lunches in the school's dining hall per month.

Solution:
Assume that:

- Half of the pupils in the school buy their lunch from the dining hall.
- On average £4 is spent on lunch each day by those pupils.

$$400 \times 4 \times 20 = 32000 \approx £30{,}000$$

You may have estimated a different proportion of pupils that use the dining hall, and a different average spend. This is okay, within reason. It is expected your final answer will vary based on your estimates of the different *variables*.

Example

Problem:
Estimate the total volume of water that a typical person drinks during their childhood.

Solution:
Assume that:

- 'Childhood' lasts 18 years (from 0 to 18).
- On average a child might drink 1 litre of water a day.

$$1 \times 365 \times 18 = 6570 \approx 6600 \text{ litres}$$